KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2023
Department of Computer Science and Engineering
CSE 4127
Image Processing and Computer Vision

TIME: 3 hours | FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the immediate right column of the questions indicate full marks.
iii) The rightmost column indicates the course outcomes.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) For the following two signals $f(x)$ and $g(x)$, find $f(x) \star g(x)$, where $\star$ is a convolution operator. (07) [CO1]

$$f(x) = \begin{cases} 1 & 0 \le x \le 1 \\ 0 & otherwise \end{cases} \qquad g(x) = \begin{cases} 1/2 & 0 \le x \le 1 \\ 0 & otherwise \end{cases}$$

b) You are assigned to smooth the following image while preserving the edges. You planned to use a bilateral filter of size 3×3, and Gaussian function for both spatial kernel and range kernel. If you use $\sigma_s = 0.6$, and $\sigma_r = 15$ for those kernels, then show the spatial, range, and bilateral kernel for the encircled position of the following image. (16) [CO2]

$$\begin{bmatrix} 10 & 12 & 76 & 70 \\ 6 & \textcircled{5} & 66 & 68 \\ 4 & 25 & 55 & 62 \\ 8 & 9 & 71 & 58 \end{bmatrix}$$

c) For the given filter, determine the DTFT $H(u, v)$ of the filter (assuming the origin is at the center), and sketch the one-dimension profiles $H(u, 0)$ and $H(0, v)$. (12) [CO3]

$$\begin{bmatrix} -1 & -3 & -1 \\ 0 & 0 & 0 \\ 1 & 3 & 1 \end{bmatrix}$$

2. a) Show that Laplacian sharpening is a special case of an Un-Sharp Mask (USM) filter. (07) [CO3]
b) Use required mathematical formulation to show that the between class variance used by Otsu's thresholding method can be written as (14) [CO1]

$$\sigma_B^2(k) = \frac{(m_G P_1(k) - m(k))^2}{P_1(k)(1 - P_1(k))}$$

c) Estimate the optical flow for the following successive video frames using Lucas–Kanade method. Use masks suggested by Horn-Schunck to compute necessary partial derivatives. Also use a 2×2 spatial coherence window, and show the estimated result for encircled position only. (14) [CO2]

$$\begin{bmatrix} 1 & 0 & 1 & 1 \\ 2 & \textcircled{9} & 9 & 8 \\ 1 & 9 & 9 & 10 \\ 0 & 9 & 8 & 9 \end{bmatrix}$$
Frame at t

$$\begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 2 & 10 & 9 \\ 1 & 1 & 9 & 8 \end{bmatrix}$$
Frame at $t + 1$

3. a) Geometric transformation of the following image is done by an inverse polynomial mapping function such as: $Y = WP(X)$, where, (13) [CO3]

$$X = [x_1 \quad x_2]^T, Y = [y_1 \quad y_2]^T,$$
$$P(x) = [1 \quad x_1 \quad x_2 \quad x_1^2 \quad x_1 x_2 \quad x_2^2], \text{ and}$$
$$W = \begin{bmatrix} 0.1 & 1 & 0 & 0.01 & -0.01 & 0.01 \\ 0.4 & 0 & 1 & 0.02 & -0.01 & 0.04 \end{bmatrix}$$

Determine output intensity at (1, 2), if bilinear interpolation is used during transformation.

x_2 \ x_1	0	1	2	3	4
0	100	105	108	109	102
1	99	104	109	100	103
2	200	204	205	102	105
3	201	106	101	103	111
4	206	208	204	200	100

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b) Let two interest points are described by some feature descriptors. Provide at least two matching ideas, with necessary equations, that can be used to decide whether those interest points are same or not. Also mention the pros and cons of those matching ideas. (12) [CO4]

c) Write Niblack's adaptive thresholding algorithm. Mention its use-cases. (10) [CO1]

4. a) During Hough transform based line detection, generally we get similar vote count in a small neighborhood. Explain what strategy can be applied to locate the peak voting position in the accumulator array. (06) [CO4]

b) Apply a 3×3 vector median filter on the location $(2, 2)$ of the given RGB image. (13) [CO2]

$$R = \begin{bmatrix} 1 & 0 & 2 & 1 \\ 3 & 2 & 1 & 2 \\ 3 & 0 & 2 & 4 \\ 2 & 1 & 0 & 1 \end{bmatrix} \quad G = \begin{bmatrix} 2 & 1 & 2 & 1 \\ 1 & 3 & 1 & 0 \\ 3 & 0 & 2 & 2 \\ 1 & 2 & 2 & 3 \end{bmatrix} \quad B = \begin{bmatrix} 3 & 3 & 1 & 2 \\ 2 & 3 & 0 & 1 \\ 1 & 0 & 2 & 1 \\ 3 & 1 & 2 & 2 \end{bmatrix}$$

c) Use DiZenzo-Cumani operator to get the edge strength and orientation for the following color image: values inside the parenthesis means (r,g,b) color vector. Also mention the importance of the operator compared to monochromatic edge detection technique for color image. Show result only for the center pixel. (16) [CO3]

$$\begin{bmatrix} (3,2,1) & (2,1,2) & (2,3,0) \\ (1,3,1) & (3,0,1) & (2,2,3) \\ (1,1,0) & (0,2,4) & (1,1,1) \end{bmatrix}$$

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

a) Define digital image processing. Briefly discuss the fundamental steps in digital image processing. (10) [CO1]

b) Consider the image segment shown below. (12) [CO2]

	3	1	2	1 ← q
	2	2	0	2
	1	2	1	1
p →	1	0	1	2

Compute length of the shortest-4, shortest-8 and shortest-m paths between pixels p and q where, $v = \{1, 2\}$. If a particular path does not exist between these two points, explain why.

c) Consider the following 3-bit image. (13) [CO3]

i) Generate the equalized image by using global histogram equalization processing.

ii) Plot the histogram of the processed image.

2	3	3	5	7
5	4	3	1	6
1	5	2	4	7
2	3	6	4	1
6	5	3	2	4

6. a) Define morphological gradient. Discuss when do we use top-hat and bottom-hat transformation in grayscale image. (10) [CO2]

b) Let, you are asked to enhance dark areas of an image while leaving the bright area as unchanged as possible. Propose strategy with necessary formulation to handle the problem. Use example to explain that the proposed method works. (14) [CO3]

c) Briefly discuss the following morphological image processing operations with necessary equations. (11) [CO3]

i) Convex hull.

ii) Pruning.

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